

Notes

January 29, 2024

Abstract

1 Selective inference II – Comparison of test statistics

Assume we only have one covariate and fix one node with $n = 50$. Let T_n^2 be defined as defined in (Shih, 2004) (or U^2 in (Hawkins, 1977)). Figure 1 shows a histogram of $-T_n^2$ (red) and the test statistic arising from our MSEF difference approach

$$\min_k Z_k = \min_k \frac{k+1}{k-1} \sum_{i=1}^k (Y_i - \bar{Y}_k)^2 + \frac{n-k+1}{n-k-1} \sum_{i=k+1}^n (Y_i - \bar{Y}_k')^2 - \frac{n+1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2$$

in blue.

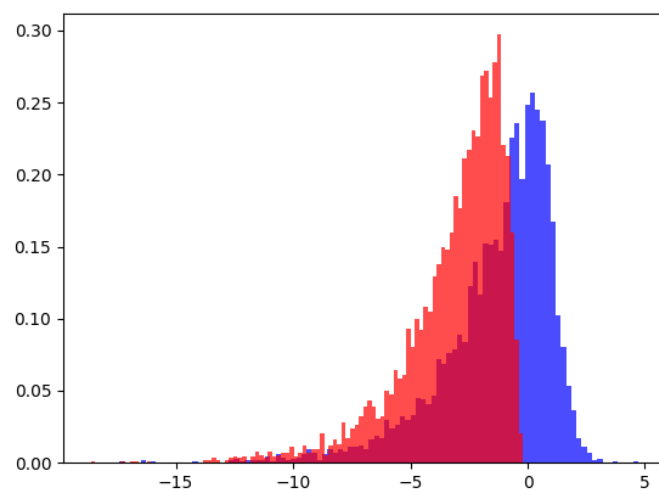


Figure 1:

Figure 2 plots $\min_k Z_k$ against $-T_n^2$.

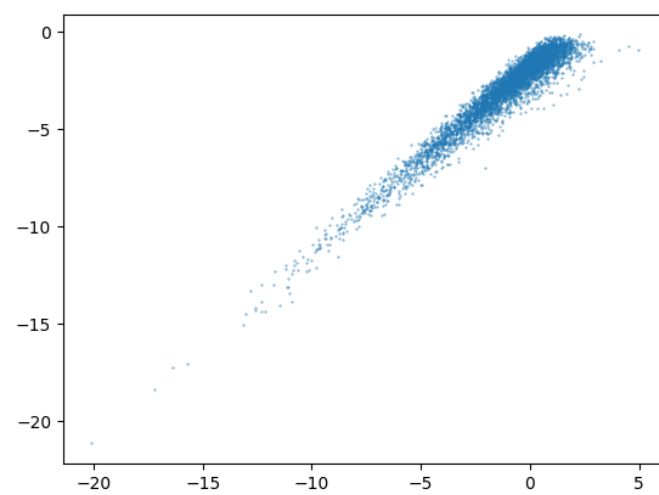


Figure 2:

Upon replacing $(n_A+1)/(n_A-1)$, $(n_B+1)/(n_B-1)$ and $(n+1)/(n-1)$ by 1, some properties of the distribution changes. E.g. $-T_n^2 \leq 0$ unlike $\min_k Z_k$. Still, there seems to be a more or less linear relationship (cf. Figure 2), and also the left tails (the important part), are similar.

Since T_n^2 is derived by a likelihood-ratio approach, one can expect that it is a better test statistic for testing H_0 against H_1 (as defined in (Hawkins, 1977)) than our MSEP-motivated one. (Better in the sense of fewer type 2 errors).

Approximations of T_n^2 (cf. Yao/Davis 1984) The next figure shows the distribution function of $-T_n^2$ (blue) and the approximation derived by Yao and Davis as given in (Shih, 2004, p. 598, eq (YD)) (red) under the null.

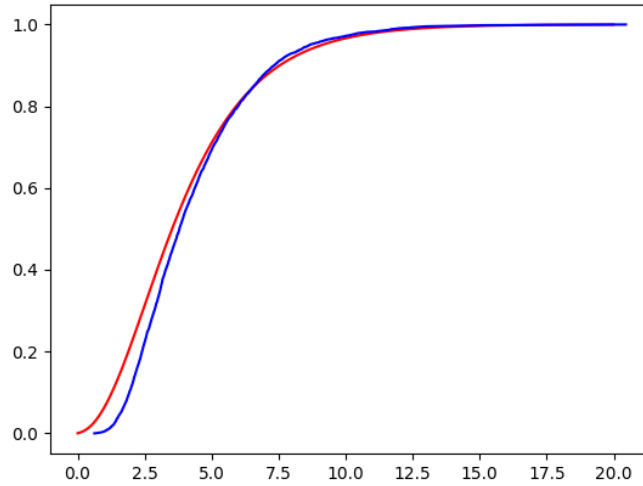


Figure 3:

The next figure shows the same, but given there is some signal, so only the distribution of T_n^2 changes (the approximation is only derived under the null).

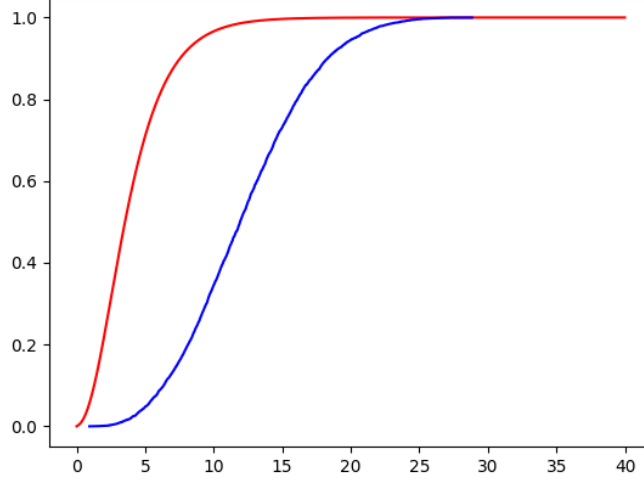


Figure 4:

The approximation under H_0 seems fine even for small n (here, still, $n = 50$).

σ^2 -dependence As argued in (Yao, Davis, 1984, Remark 2.4), if σ^2 is unknown, then with $\hat{\sigma}^2 = 1/n \sum (Y_i - \bar{Y})^2$,

$$\tilde{T}_n^2 = T_n^2 / \hat{\sigma}^2$$

asymptotically as $n \rightarrow \infty$ has the same distribution as T_n^2 given that $\sigma^2 = 1$ (also the approximation above is under the assumption that $\sigma^2 = 1$). We can hence compute $\tilde{T}_n^2$ and plug it into the cdf approximation (YD) to get a selected-inference adjusted p-value.

Minimal number of data points per leaf m_{min} Both the exact distribution of T_n^2 as derived in (Hawkins, 1977), as well as its approximation by (Yao, Davis) assume that $m_{min} = 0$. Setting m_{min} to a larger value can therefore distort the distributions since fewer splitpoints are allowed. We have to try out which values of m_{min} still work well.